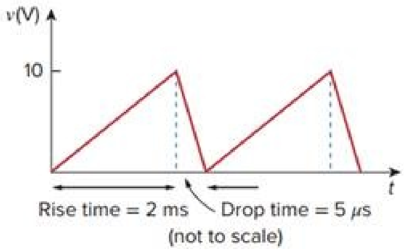
Problem

To move a spot of a cathode-ray tube across the screen requires a linear increase in the voltage across the deflection plates, as shown in Fig. 7.151. Given that the capacitance of the plates is 4 nF, sketch the current flowing through the plates.

Fig. 7.151:



Step-by-step solution

Step 1 of 2

$$i = C\frac{dV}{dt} = 4 \times 10^{-9} \begin{cases} \dfrac{10}{2\times10^{-3}} & 0 < t < t_r \\ \dfrac{-10}{5\times10^{-6}} & t_R < t < t_0 \end{cases}$$

$$i(t) = \begin{cases} 20\,\mu A & 0 < t < 2mS \\ -8mA & 2mS < t < 2mS + 8\,\mu S \end{cases}$$

Step 2 of 2

Which is sketched below